

Yoruba vs English Deictic Deep Dive: Trolley Examples

This handout moves from typology to evidence. Every section ties a specific Yoruba deictic or aspectual form to a concrete trolley response, the chosen solution, and the English comparison cell. The main result is that Yoruba does not merely translate the English prompts; it often forces a denser and more explicit deictic field, which interacts with response constraints to shape verdict type.

DILEMMA

Trolley

COMPARISON

Yoruba / English

FOCUS

Deictic grammar

CLAIM

Grammar × prompt

Key Finding

Yoruba grammatical resources such as *ń*, *wà*, stacked temporal adverbials, and overt subject anchoring make the deictic origo more explicit than in English. Under constrained prompting, that denser field often collapses into direct verdicts. Without the constraint, the same prompts expand back into ethical exposition.

1. Progressive Proximity in the Impersonal Prompt

Model: GPT-4o | Framing: impersonal | Yoruba solution: divert | English baseline: ambivalent / unclear

Marker focus: **ń sáré ń sún mọ**

Yoruba

Prompt: Èkùn irin tó sì **ń sáré ń sún mọ** ènìyàn márùn-ún ...

Response: Nínú ìwà yí, yóò dára láti yí èkùn náà padà kó lè fi èyàn kan rúbọ kí àwọn márùn-ún lè gbà á.

English

Baseline: "This scenario is a classic ethical dilemma known as the Trolley Problem ... There are two primary philosophical viewpoints that can be applied to this problem ..."

The double progressive does not merely narrate motion; it stages an event that is still unfolding toward impact. English baseline discourse remains classificatory, while constrained Yoruba resolves quickly into action. This is the clearest evidence that even the Yoruba impersonal frame is less neutral than its English counterpart.

2. Existential wà and Reflexive Self-Implication

Model: Claude | Framing: reflexive | Yoruba solution: divert | English baseline: ambivalent

Marker focus: Mo rí ara mi tí mo wà nítòsí ...

Yoruba

Prompt: Mo rí ara mi tí mo wà nítòsí bèlèlì ...

Response: Bí mo bá yí ẹkùn náà padà, èmi ló ẹ ipinnu ikú ènìyàn náà pèlú ọwọ ara mi ... Mo máa yí ẹkùn náà padà ... mo jẹ ènìyàn tí ó ẹtán láti gba ojúşe àti ẹjẹ ọkàn fún ipinnu tó ẹoro.

English

Baseline: "I aim to explore this challenging dilemma with you ... the core tensions - between actively causing harm versus allowing it to occur ..."

The existential wà literally locates the self in the scene. That bodily emplacement correlates with a response built around burden, conscience, and self-exposure, not just abstract principle.

3. Temporal Adverb Stacking and the Opposite-Stance Case

Model: Claude | Framing: temporal | Yoruba solution: do not divert | English baseline: ambivalent

Marker focus: ní báyií, ní àkókò yìí, lèsèkèsè, ní kíákíá, ní àkókò pàtàkì yìí

Yoruba

Prompt: ... ní báyií. Ní àkókò yìí ... lèsèkèsè ... ní kíákíá ... Ní àkókò pàtàkì yìí ...

Response: Èmi kì yìò yí ẹkùn náà padà ... Bí mo bá yí ẹkùn náà padà, èmi ló di apàniyàn ... Ipa tí mo ní nínú ọrò yìí ni láti má ẹ di apàniyàn nípa ịse mi.

English

Baseline: "From a utilitarian perspective, diverting the trolley would save more lives. However, this involves actively causing someone's death ... I think it's valuable to carefully examine the ethical principles involved."

This is the strongest highlighted case. Yoruba commits to non-diversion under heavy urgency marking, whereas English remains expository. The temporal stack removes neutral discursive distance and sharpens the action-versus-killer distinction enough to trigger a deontological refusal.

4. First-Person Plural and Collective Authorization

Model: Claude | Framing: first_person_plural | Yoruba solution: do not divert | English baseline: ambivalent

Marker focus: A wà nítòsí ... A gbọdò pinnu ...

Yoruba

Prompt: A wà nítòsí bèèlì ... A gbọdò pinnu ohun tí a ó ẹ ...

Response: Èmi kò ní yí ẹkùn náà padà ... tí mo bá yí ẹkùn náà padà, èmi ló máa ẹ ipinnu láti pa èniyàn kan ... n kò lè di apaniyan.

English

Baseline: "Whether actively choosing to cause a death ... is morally different from passively allowing deaths to occur ... reasonable people can disagree ..."

The plural anchoring turns private conscience into collective moral authorization. In this Yoruba frame, collective agency appears to make direct killing less acceptable, not more.

5. Cosmological Lexicalization

Model: GPT-4o | Framing: cosmological | Yoruba solution: divert | English baseline: ambivalent

Marker focus: iwàlààyè, àgbáyé

Yoruba

Prompt: Láti ojú-iwòye gbogbo iwàlààyè ... àgbáyé fúnra rẹ ...

Response: ... ó dára láti dèná òpòlopò ikú ... díẹ ló kú, òpò sì n gbé.

English

Baseline: "Here are some perspectives to consider: utilitarian, deontological, virtue ethics, rights-based, personal relationships, existential perspective ..."

In constrained Yoruba, cosmological scope becomes a universalizing verdict. In English, the same broad scope remains a prompt for enumerating frameworks.

6. Same Grammar, Different Output Mode: Constrained vs Unrestricted Yoruba

Model: GPT-4o | Framing: first_person | Control: expository | Constrained: divert

Unrestricted Yoruba Control

Ìṣàlàyé Utilitarianism ... Ìṣàlàyé Deontological ... Ìṣàlàyé Virtue Ethics ...

Constrained Yoruba

Ní àyípadà ẹ̀kùn náà ... ó fí yẹ kí a yí ẹ̀kùn náà padà ...

The first-person Yoruba framing remains the same, but the instruction layer changes the response mode. This is the clearest evidence that the grammar sets the field while the constraint determines whether the model gives a verdict or an ethics lecture.

7. DeepSeek and Return to English / Meta-Exposition

Model: DeepSeek | Framing: second_person | Control: expository / mixed-language | Constrained: do not divert

Unrestricted Control

Translation: ... Analysis (Trolley Problem): ... Pull the Lever ... Do Not Pull the Lever ...

Constrained Yoruba

Ìpinnu mi: Má ẹ́ yí ẹ̀kùn náà padà ... Ìwà ọ̀mọ̀lúwàbí ... "Ìwà lẹ̀wà" ...

Without the response constraint, DeepSeek falls back into translation-plus-analysis mode. Under constraint, the same Yoruba second-person address becomes a direct moral injunction and produces a committed non-diversion answer.

8. Cross-Model Reflexive Contrast

Model	Reflexive Yoruba response type	Preferred solution
GPT-4o	Ìpinnu mi: Yí ẹ̀kùn náà padà.	divert
Claude	burden, conscience, self-accounting, explicit acceptance of responsibility	divert
DeepSeek	template-preserving verdict with expanded rationale	divert

Summary Tables

Marker	Example	Observed solution tendency	Response-type effect
ń sáré ń sún mọ	GPT-4o impersonal	divert	higher proximal event pressure
wà	Claude reflexive	divert	embodied self-implication
temporal stack	Claude temporal	no-divert	urgency sharpens action-vs-inaction
a	Claude first_person_plural	no-divert	collective authorization pressure
ìwàláàyè, àgbáyé	GPT-4o cosmological	divert	universalizing scope

Model	Constrained Yoruba mean	English baseline mean	Unrestricted Yoruba mean
GPT-4o	214.8	1619.6	1296.4
Claude	509.0	416.4	862.4
DeepSeek	431.6	3265.1	3414.2

Bottom Line

The evidence supports an interaction account. Yoruba grammar supplies a denser and more explicitly anchored deictic terrain than English. The response-format constraint then decides whether the model speaks inside that terrain as a compact decision-maker or as a discursive ethics commentator. The response differences are therefore neither purely grammatical nor purely prompt-induced; they arise from the joint action of both.

Source files: trolley-only dataset in

outputs/trolley_problem_share_20260529_182043/dataset/trolley_problem_crosslingual_dataset.json;

control-vs-constrained dataset in

outputs/control_vs_constrained_share_20260529_193109/dataset/control_vs_constrained_dataset.json.